

Two Port Networks- z-Parameter, y-Parameter	
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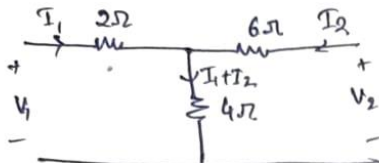
Aim:

To verify Two Port Networks- z-Parameter, y-Parameter.

Theoretical Calculation:

Que 1. Calculate Z-parameter for the given Network.

Que 1 find Z-parameter



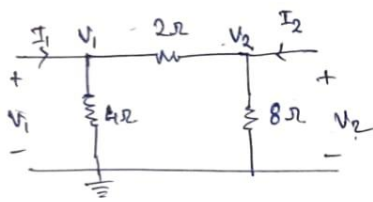
$$Z = \begin{bmatrix} 6 & 4 \\ 4 & 10 \end{bmatrix} \Omega$$

$$\begin{aligned} V_1 &= 2I_1 + 4(I_1 + I_2) \\ V_1 &= 6I_1 + 4I_2 \\ V_1 &= Z_{11}I_1 + Z_{12}I_2 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} Z_{11} = 6\Omega \\ Z_{12} = 4\Omega \end{array}$$

$$\begin{aligned} V_2 &= 6I_2 + 4(I_1 + I_2) \\ V_2 &= 10I_2 + 4I_1 \\ V_2 &= 4I_1 + 10I_2 \\ V_2 &= Z_{21}I_1 + Z_{22}I_2 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} Z_{21} = 4\Omega \\ Z_{22} = 10\Omega \end{array}$$

Que 2. Calculate Z-parameter for the given Network.

Que 2 Calculate the Z parameter:



KCL at V_1

$$I_1 = \frac{V_1}{4} + \frac{V_1 - V_2}{2}$$

$$8I_1 = 2V_1 + 4V_1 - 4V_2$$

$$8I_1 = 6V_1 - 4V_2 \quad \text{--- (1)}$$

KCL at V_2

$$I_2 = \frac{V_2}{8} + \frac{V_2 - V_1}{2}$$

$$16I_2 = 2V_2 + 8V_2 - 8V_1$$

$$16I_2 = 10V_2 - 8V_1 \quad \text{--- (2)}$$

Put eq (2) and (1)

$$\Rightarrow V_2 = \frac{16I_2 + 8V_1}{10} \quad \text{--- (2)}$$

$$6V_1 = 4V_2 - 8I_1$$

$$6V_1 = 4 \left(\frac{16I_2 + 8V_1}{10} \right) - 8I_1$$

$$V_1 = \frac{4}{60} (16I_2 + 8V_1) - \frac{8I_1}{6}$$

$$\Rightarrow V_1 = \frac{16I_2}{15} + \frac{32V_1}{60} - \frac{8I_1}{6}$$

$$\Rightarrow V_1 - \frac{32}{60}V_1 = \frac{16I_2}{15} - \frac{8I_1}{6}$$

$$\Rightarrow \frac{7V_1}{15} = \frac{16I_2}{15} - \frac{8I_1}{6}$$

$$\Rightarrow V_1 = \frac{16I_2}{7} - \frac{8I_1}{42}$$

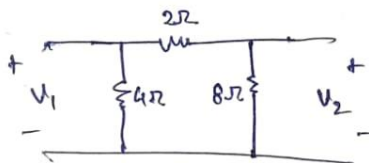
$$\Rightarrow V_1 = \frac{8 \cdot 15}{42} I_1 - \frac{16}{7} I_2$$

$$\Rightarrow \frac{60}{21} I_1 - \frac{16}{7} I_2$$

$$\Rightarrow 2.85 I_1 - 2.28 I_2 = V_1 \quad \text{--- (3)}$$

$$Z_{11} I_1 = Z_{12} I_2 = V_1$$

Que 2



$$Z_{11} = 2.85 \Omega, \quad Z_{12} = 2.28 \Omega$$

Put eq (3) in (1)

$$8I_1 = 6(2.85I_1 - 2.28I_2) - 4V_2$$

$$V_2 = \frac{(-8I_1 + 17.1I_1 - 13.7I_2)}{4}$$

$$Z_{21} = 2.28 \Omega$$

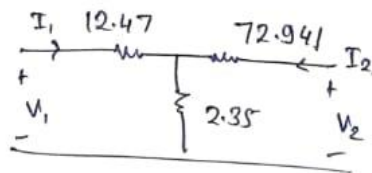
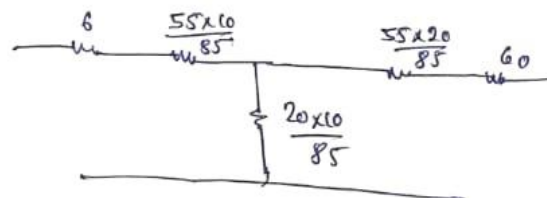
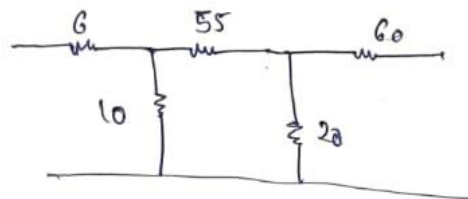
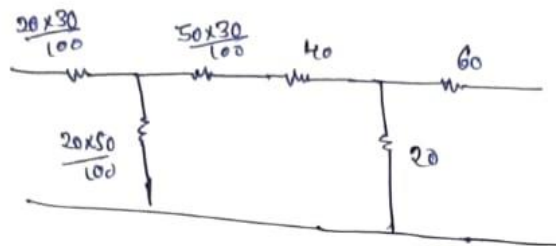
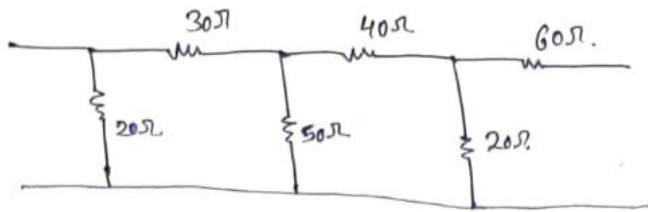
$$Z_{22} = 3.43 \Omega$$

$$\begin{cases} V_2 = 2.28 I_1 - 3.43 I_2 \\ V_2 = Z_{21} I_1 - Z_{22} I_2 \end{cases}$$

$$Z = \begin{bmatrix} 2.85 & 2.28 \\ 2.28 & 3.43 \end{bmatrix} \Omega$$

Que 3. Calculate Z-parameter for the given Network.

Que 3 Calculate the Z parameter of the given Network



$$Z = \begin{bmatrix} 14.82 & 2.35 \\ 2.35 & 75.26 \end{bmatrix} \Omega$$

$$\left. \begin{aligned} V_1 &= 14.82 I_1 + 2.35 I_2 \\ V_1 &= Z_{11} I_1 + Z_{12} I_2 \end{aligned} \right\} \begin{aligned} Z_{11} &= 14.82 \Omega \\ Z_{12} &= 2.35 \Omega \end{aligned}$$

$$\left. \begin{aligned} V_2 &= 2.35 I_1 + 75.26 I_2 \\ V_2 &= Z_{21} I_1 + Z_{22} I_2 \end{aligned} \right\} \begin{aligned} Z_{21} &= 2.35 \Omega \\ Z_{22} &= 75.26 \Omega \end{aligned}$$

Que 4. Calculate Y-parameter for the given Network.

Que 4

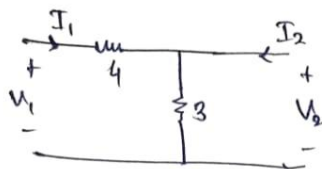
$$[Z]^{-1} = Y = \begin{bmatrix} 14.82 & 2.35 \\ 2.35 & 76.26 \end{bmatrix}^{-1} \text{ ohm}$$

$$Y = \begin{bmatrix} 0.0678 & 0.00211 \\ 0.00211 & 0.01335 \end{bmatrix} \text{ ohm}^{-1} \text{ or mho}$$

Que 5. Calculate Y-parameter for the given Network.

Que 5

Calculate y parameter



$$Z = \begin{bmatrix} 7 & 3 \\ 3 & 3 \end{bmatrix} \Omega$$

$$Z^{-1} = Y = \begin{bmatrix} 0.25 & -0.25 \\ -0.25 & 0.5833 \end{bmatrix} (\text{ohm})^{-1}$$

$$V_1 = -4I_1 + 3(I_1 + I_2)$$

$$\begin{aligned} V_1 &= -4I_1 + 3I_1 + 3I_2 \\ V_1 &= -I_1 + 3I_2 \end{aligned} \quad \left. \begin{aligned} V_1 &= Z_{11}I_1 + Z_{12}I_2 \\ V_1 &= -I_1 + 3I_2 \end{aligned} \right\} \begin{aligned} Z_{11} &= -1\Omega \\ Z_{12} &= 3\Omega \end{aligned}$$

$$V_2 = 0 + 3(I_1 + I_2)$$

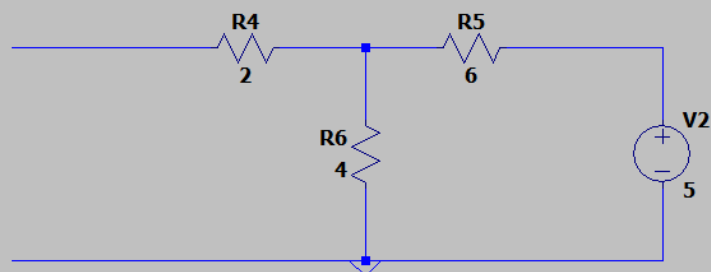
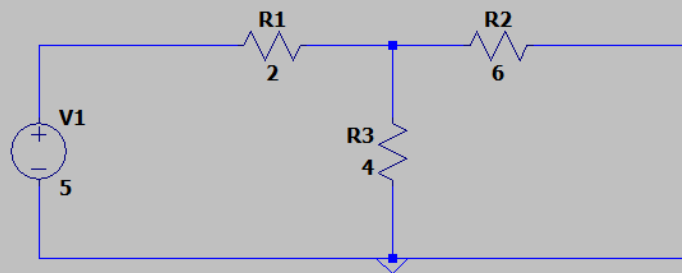
$$\begin{aligned} V_2 &= 3I_1 + 3I_2 \\ V_2 &= Z_{21}I_1 + Z_{22}I_2 \end{aligned} \quad \left. \begin{aligned} V_2 &= 3I_1 + 3I_2 \\ V_2 &= Z_{21}I_1 + Z_{22}I_2 \end{aligned} \right\} \begin{aligned} Z_{21} &= 3\Omega \\ Z_{22} &= 3\Omega \end{aligned}$$

Verification by Simulation:

Que 1. Calculate Z-parameter for the given Network.

Circuit Diagram:

Que 1. Calculate the z-parameter of the given network.



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Plot of Voltage/Current:

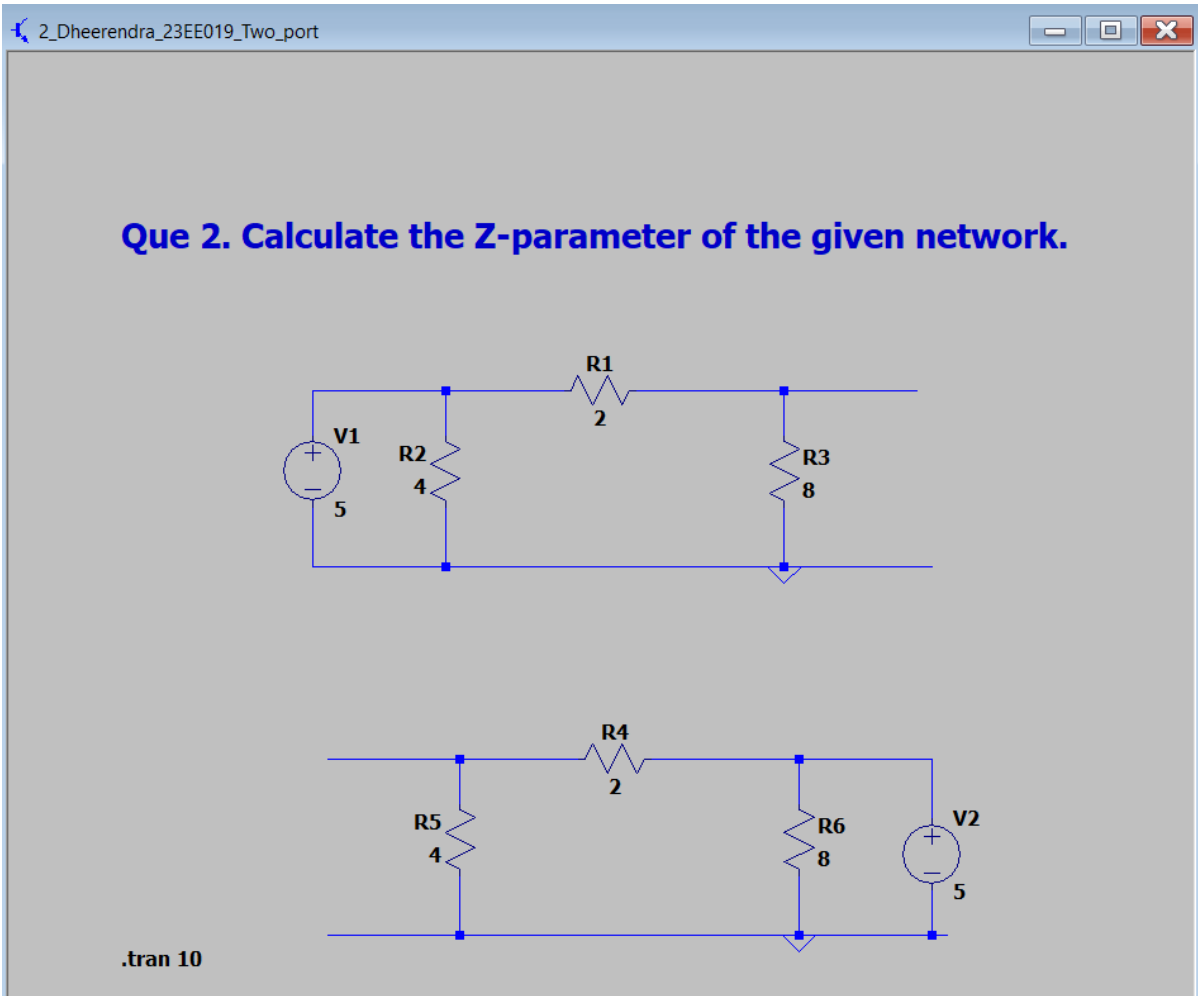


Observation Table:

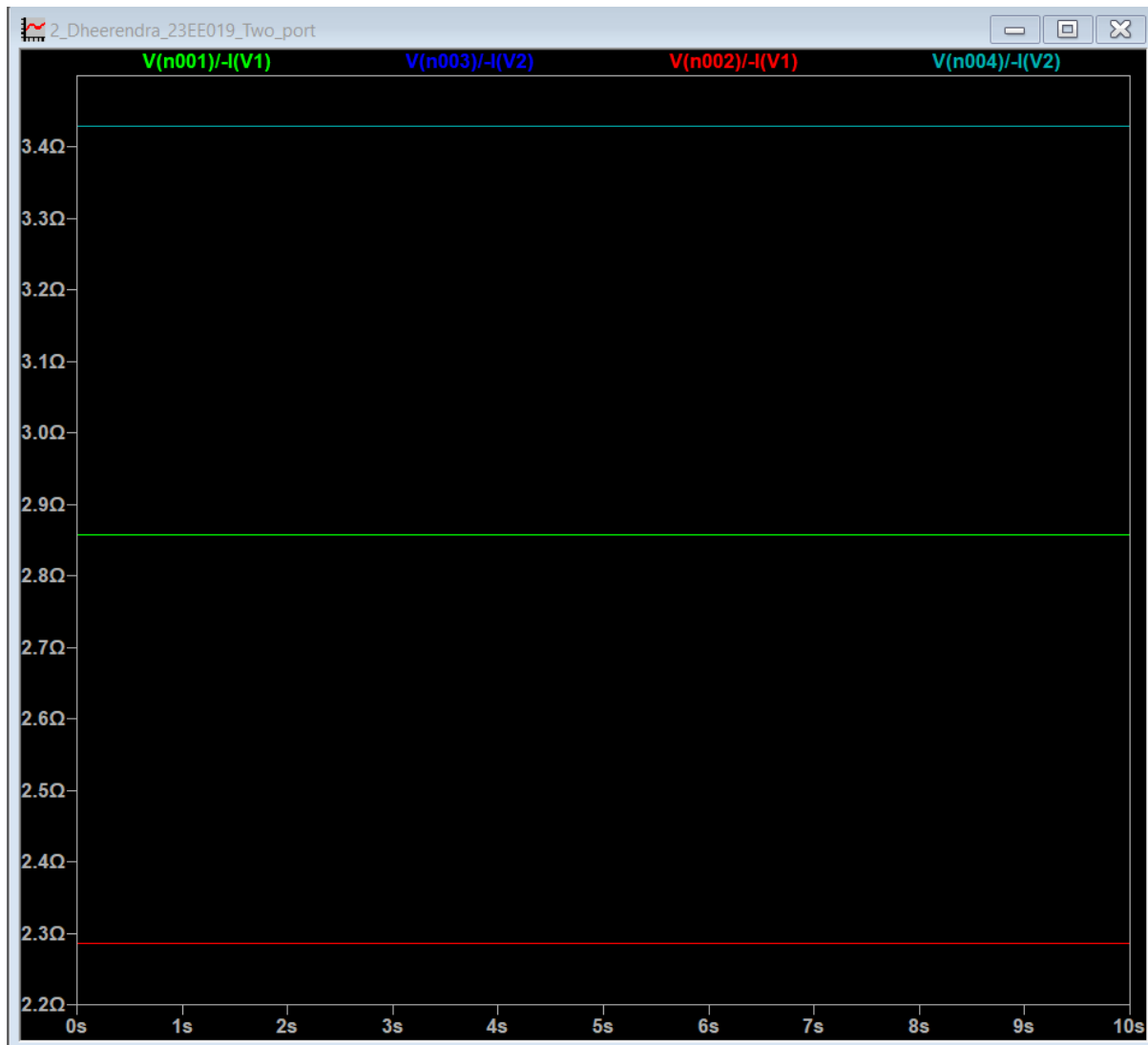
S. No.	z-Parameters	Theoretical Value	Practical Value
1	Z_{11}	6 Ohm	6.0000001Ohm
2	Z_{12}	4 Ohm	4 Ohm
3	Z_{21}	4 Ohm	4 Ohm
4	Z_{22}	10 Ohm	10 Ohm

Que 2. Calculate Z-parameter for the given Network.

Circuit Diagram:



Plot of Voltage/Current:



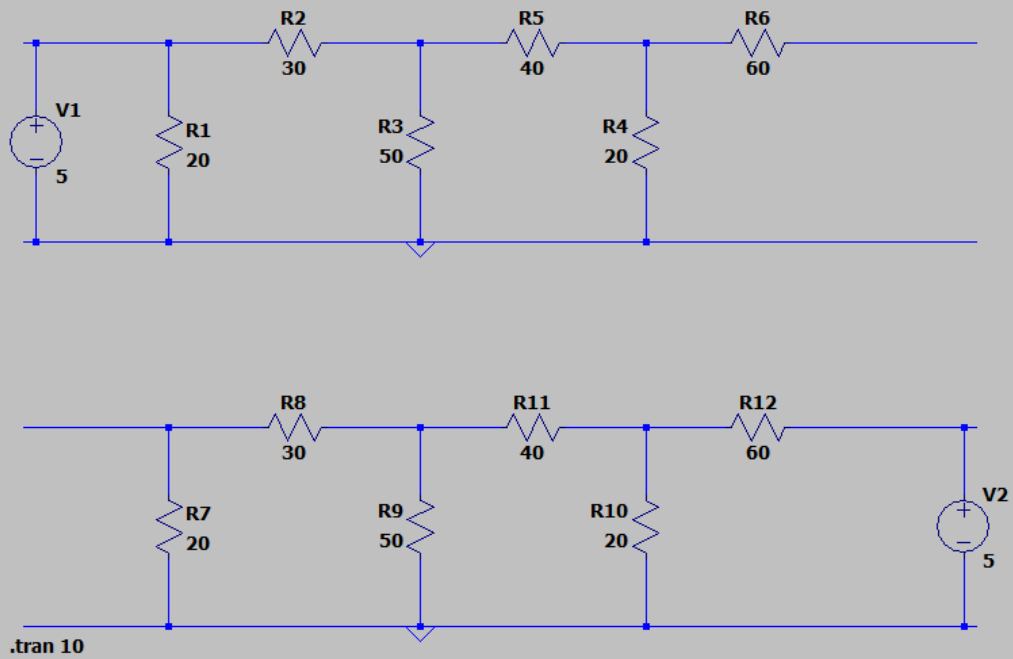
Observation Table:

S. No.	z-Parameters	Theoretical Value	Practical Value
1	Z_{11}	2.85 Ohm	2.8571429 Ohm
2	Z_{12}	2.28 Ohm	2.2857142 Ohm
3	Z_{21}	2.28 Ohm	2.2857143 Ohm
4	Z_{22}	3.43 Ohm	3.4285713 Ohm

Que 3. Calculate Z-parameter for the given Network.

Circuit Diagram:

Que 3. Calculate the Z-parameter of the given network



Plot of Voltage/Current:



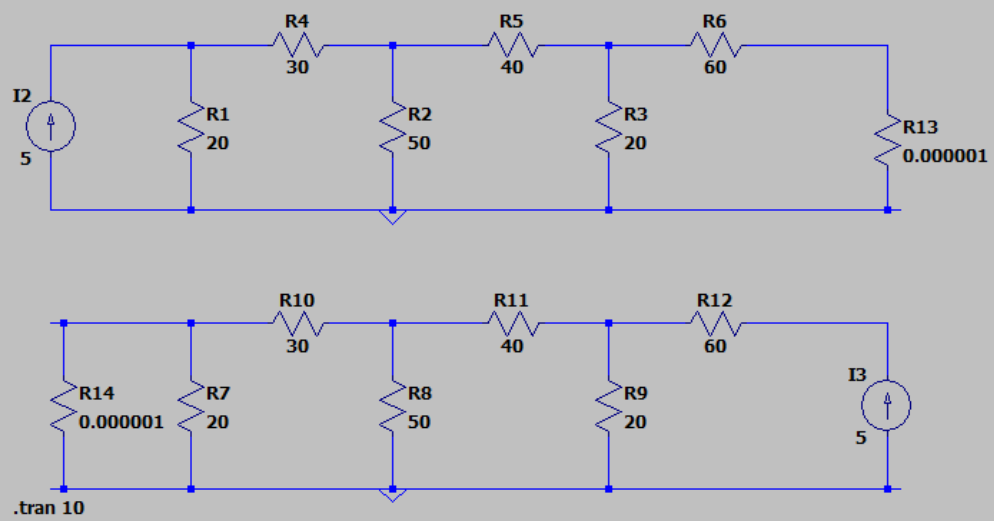
Observation Table:

S. No.	z-Parameters	Theoretical Value	Practical Value
1	Z_{11}	14.82 Ohm	14.82353Ohm
2	Z_{12}	2.35 Ohm	2.3529412Ohm
3	Z_{21}	2.35 Ohm	2.3529412Ohm
4	Z_{22}	75.26 Ohm	75.294118Ohm

Que 4. Calculate Y-parameter for the given Network.

Circuit Diagram:

Que 4. Calculate the Y- parameters of the given Network.



Plot of Voltage/Current:

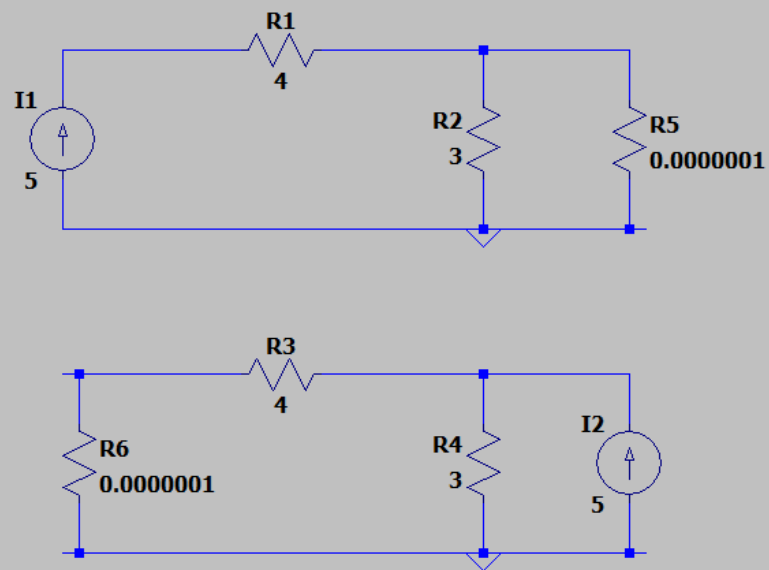


Observation Table:

S. No.	y-Parameters	Theoretical Value	Practical Value
1	Y_{11}	67.8 mOhm-1	67.79661mOhm-1
2	Y_{12}	2.11 mOhm-1	2.1186439mOhm-1
3	Y_{21}	2.11 mOhm-1	2.1186441mOhm-1
4	Y_{22}	13.35 mOhm-1	13.347457mOhm-1

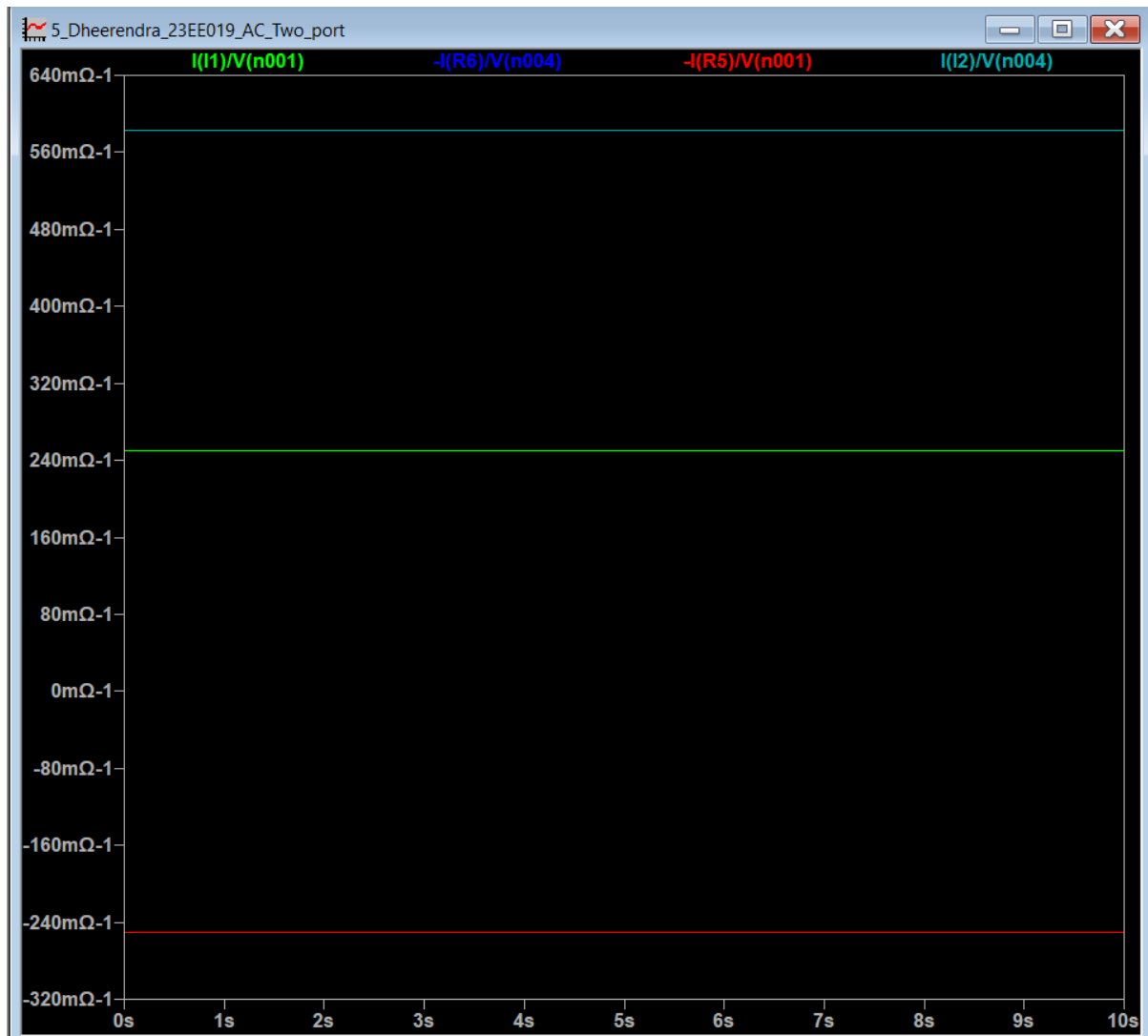
Que 5. Calculate Y-parameter for the given Network.

Circuit Diagram:

Que5. Calculate Y-parameters of the given Network

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Plot of Voltage/Current:



Observation Table:

S. No.	y-Parameters	Theoretical Value	Practical Value
1	Y_{11}	0.25 mho	250mOhm-1
2	Y_{12}	0.25 mho	-250mOhm-1
3	Y_{21}	0.25 mho	-250mOhm-1
4	Y_{22}	0.583 mho	583.33335mOhm-1

Results and Conclusion:

As we see that the theoretical values of the Z parameters and the Y parameters is approximately equal to the practical values of the Z parameters and the Y parameters, Hence the Z parameters and Y parameters are Verified.